

HEMARUKA AGCM, AB, Canada

WMO: 717930

Lat: 51.7779N

Lon: 111.2119W

Elev: 2559

StdP: 13.39

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.1	-20.3	-32.4	1.0	-26.0	-26.3	1.4	-20.4	25.5	3.8	22.0	0.5	4.5	190	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.1	86.1	61.4	82.2	61.4	78.7	59.9	65.4	77.1	63.5	75.8	62.0	74.5	11.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	89.4	70.3	59.3	82.7	68.3	57.4	77.2	66.2	31.6	76.7	30.2	76.1	29.0	74.3	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.2	23.2	20.0	DB	-33.1	91.8	4.1	5.0	-36.1	95.4	-38.5	98.3	-40.8	101.1	-43.7	104.7	(4)
(5)			WB	-33.3	69.4	4.1	2.1	-36.2	70.9	-38.6	72.1	-40.9	73.3	-43.8	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	12.3	9.7	23.2	37.3	50.5	57.9	63.2	61.8	52.5	38.9	23.4	10.2		
	DBStd	22.38	14.53	14.33	14.39	9.60	7.65	5.88	5.32	6.07	8.34	9.23	12.99	14.87		
	HDD50	6075	1170	1129	831	389	90	8	0	2	67	358	799	1232		
	HDD65	10356	1635	1549	1296	830	451	224	99	137	381	808	1249	1697		
	CDD50	1292	0	0	0	9	105	245	408	366	142	16	1	0		
	CDD65	96	0	0	0	0	1	10	42	36	7	0	0	0		
	CDH74	2049	0	0	0	6	96	223	750	700	267	7	0	0		
	CDH80	577	0	0	0	0	10	46	225	211	85	0	0	0		
	WSAvg	9.5	8.8	8.2	9.8	10.6	10.2	10.0	9.5	9.1	9.0	10.7	8.8	8.8		
	PrecAvg	14.0	0.6	0.4	0.6	0.9	1.6	2.9	2.3	1.9	1.1	0.6	0.6	0.5		
(15) Precipitation	PrecMax	18.3	1.7	0.9	1.4	2.0	3.3	6.0	5.8	3.4	2.9	1.3	2.1	0.7		
	PrecMin	10.4	0.1	0.0	0.2	0.1	0.2	0.6	0.9	0.4	0.1	0.2	0.0	0.1		
	PrecStd	2.6	0.4	0.2	0.3	0.5	0.9	1.3	1.1	1.0	0.7	0.3	0.5	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	42.4	58.9	71.5	80.9	86.1	90.6	90.8	87.7	73.0	58.8	42.9		
		MCWB	37.3	36.9	46.0	49.3	56.4	62.0	63.9	61.5	59.0	53.8	45.0	37.7		
	2%	DB	39.3	37.3	52.3	66.1	76.6	79.7	85.9	85.3	81.7	65.8	50.4	38.7		
		MCWB	35.3	33.7	41.6	48.3	54.4	59.6	62.9	61.4	58.2	50.8	41.1	34.5		
	5%	DB	36.1	33.9	46.5	60.2	72.9	75.8	81.7	81.3	75.8	60.2	44.2	34.4		
		MCWB	32.9	31.3	38.5	44.2	52.9	58.4	62.7	61.0	56.5	47.6	37.9	30.7		
	10%	DB	32.5	30.5	40.6	54.8	68.1	72.2	77.8	77.6	70.2	54.6	39.5	29.4		
		MCWB	30.1	28.3	35.2	41.9	50.9	57.1	61.4	60.1	54.4	44.8	34.8	27.1		
	0.4%	WB	37.8	37.3	46.6	51.5	58.2	65.1	69.7	67.3	61.8	55.9	46.8	38.3		
		MCDB	42.6	42.2	58.3	68.9	76.6	76.8	84.6	78.4	80.3	71.3	58.0	41.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.3	33.8	42.2	48.1	56.0	62.4	66.2	64.7	59.4	52.0	42.0	34.5		
		MCDB	38.7	36.8	50.7	65.0	72.7	75.4	77.2	76.8	76.9	61.5	49.6	38.5		
	5%	WB	33.2	31.4	38.9	45.2	54.3	60.5	64.3	62.9	57.3	48.5	38.6	31.1		
		MCDB	36.1	34.0	46.1	58.0	69.8	72.1	76.0	76.1	72.8	58.8	43.7	34.3		
	10%	WB	30.1	28.3	35.5	42.8	52.2	58.6	62.7	61.2	55.2	45.5	35.5	27.2		
		MCDB	32.5	30.4	40.1	53.8	65.9	69.2	74.5	74.4	69.7	54.2	39.0	29.4		
	5% DB	MDBR	18.9	19.6	19.6	22.8	25.4	23.2	25.1	25.7	26.4	23.2	18.9	18.0		
		MCDBR	19.7	21.6	25.8	33.1	32.9	29.4	32.4	33.4	36.8	31.9	24.7	22.4		
		MCWBR	16.2	18.1	16.0	16.7	15.0	13.0	13.0	13.3	16.2	17.7	16.5	18.1		
		MCWBR	19.6	21.1	24.9	30.3	28.9	25.1	26.4	27.8	33.8	30.0	23.3	21.6		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.321	0.289	0.343	0.367	0.375	0.362	0.380	0.324	0.285	0.259	0.258		
		taud	2.310	2.163	2.453	2.318	2.297	2.324	2.397	2.315	2.451	2.536	2.477	2.306		
	Ebn at Noon		238	247	292	284	281	279	281	268	274	268	246	221		
		Edh at Noon	20	31	28	36	39	38	35	36	28	21	16	17		
(44) All-Sky Solar Radiation	RadAvg		365	702	1144	1531	1850	1914	2046	1666	1190	719	377	275		
	RadStd		34	59	87	129	124	121	90	126	112	91	33	26		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page